

Statistical Analysis of Healthy Lifestyle Factors and Their Impact on Health Indicators

ABSTRACT

This project presents a comprehensive statistical analysis of the Healthy Lifestyle dataset using the R programming language. The study aims to examine the relationships between daily habits, demographic characteristics, and key health indicators such as blood pressure, heart rate, sleep duration, BMI, and stress levels. The analysis begins with data cleaning, including the detection and treatment of missing values, followed by the derivation of descriptive statistics for numerical and categorical variables. Using R, various graphical methods—such as histograms, boxplots, scatterplots, and correlation visualizations—are employed to identify data distributions, skewness, and potential outliers. The project further investigates the influence of explanatory variables (age, stress level, and physical activity level) on blood pressure through linear regression modeling. Additionally, the correlation between heart rate and factors such as gender, occupation, daily steps, and age is evaluated and visualized. Probabilities related to BMI categories are computed using conditional probability techniques. Finally, the study assesses the presence and effect of outliers in sleep duration and explores the relationship between sleep disorders and key health variables. Overall, the project demonstrates the use of R as a powerful tool for statistical computation, visualization, and data-driven health insights.

Keywords: R programming; Statistical modeling; Descriptive statistics; Correlation analysis; Regression analysis; Health data; BMI classification

1. Introduction

Lifestyle behaviors play a central role in shaping individual health outcomes, influencing factors such as blood pressure, sleep quality, stress, and cardiovascular performance. With the increasing availability of health-related datasets, statistical analysis has become a powerful approach for understanding how daily habits contribute to overall well-being. Analyzing such data allows researchers to detect trends, explore associations, and identify variables that significantly affect physiological measures (Ashbaugh & Zhang, 2024).

In this research, we examine the *Healthy Lifestyle* dataset using the R programming language, a widely used tool for statistical computation and data visualization. R provides efficient methods for handling missing values, summarizing variables, modeling relationships, and creating high-quality plots that reveal underlying patterns in the data (Kim & Alvarez, 2023). The dataset includes variables related to age, occupation, sleep duration, physical activity, stress level, blood pressure, BMI classification, and heart rate, enabling a comprehensive assessment of how lifestyle factors interact with measurable health indicators.

The analysis includes descriptive statistics, distribution assessment, correlation exploration, regression modeling, and conditional probability evaluation. These techniques help quantify the influence of age, stress, and physical activity on blood pressure, as well as the relationship between heart rate and variables such as gender, occupation, daily steps, and age. Through this work, the project demonstrates how statistical methods can uncover meaningful insights from real-world health data, supporting better understanding of lifestyle–health dynamics.

2. Body section

2.1 Data

The *Healthy Lifestyle* dataset used in this analysis contains **374 samples** and **13 attributes**, representing demographic information, lifestyle behaviors, and health indicators. The dataset includes a mix of numerical and categorical variables that allow for comprehensive statistical exploration.

The attributes are as follows:

- **Person ID (integer):** Unique identifier for each individual.
- **Gender (categorical):** Male or Female.
- **Age (integer):** Age of the individual in years.
- **Occupation (categorical):** Professional category such as Doctor, Engineer, Nurse, etc.
- **Sleep Duration (numeric, float):** Average hours of sleep per day.
- **Quality of Sleep (integer):** Self-rated sleep quality on a numerical scale.

- **Physical Activity Level (integer):** Level of daily physical activity measured numerically.
- **Stress Level (integer):** Self-reported stress score.
- **BMI Category (categorical):** Body mass classification such as Normal, Overweight, or Obese.
- **Blood Pressure (categorical/string):** Combined systolic/diastolic blood pressure in “####/####” format.
- **Heart Rate (integer):** Resting heart rate measured in beats per minute.
- **Daily Steps (integer):** Number of steps walked per day.
- **Sleep Disorder (categorical):** Presence or absence of conditions such as Insomnia or Sleep Apnea.

The dataset contains both continuous variables (e.g., Sleep Duration, Age, Heart Rate) and categorical variables (e.g., Gender, Occupation, BMI Category), allowing for descriptive, inferential, and predictive statistical analysis.

2.2 Methods

The project was implemented entirely in **R** using several data science, statistical, and visualization libraries, including **dplyr**, **ggplot2**, **reshape2**, **e1071**, and base R functions. The analysis followed a structured workflow consisting of the following steps:

1. **Data Cleaning and Preprocessing:** The analysis began with importing the *healthy_lifestyle.csv* dataset into R using the `read.csv()` function. A structured cleaning process was carried out to ensure data integrity, starting with the identification of explicit and hidden missing values. Empty or whitespace-only strings within character variables were converted into proper NA values, after which the dataset was rechecked to confirm that all missing entries were accurately captured. Because complete data were required for statistical modeling, all rows containing any missing values were removed using `na.omit()`, resulting in a final dataset consisting entirely of complete cases and suitable for analysis.
2. **Descriptive Statistics Computation:** Descriptive statistics were then computed to summarize the key characteristics of the dataset. Numerical variables such

as *Age*, *Sleep.Duration*, and *Heart.Rate* were evaluated using measures of central tendency and dispersion, including the mean, median, quartiles, standard deviation, and skewness. Categorical variables, including *Gender* and *Occupation*, were summarized using frequency tables and mode calculations to identify the most common categories. These descriptive measures provided an initial understanding of the overall distribution of demographic and lifestyle characteristics within the dataset.

3. **Exploratory Data Analysis and Statistical Modeling:** Exploratory Data Analysis (EDA) was conducted using histograms and boxplots to examine variable distributions and detect outliers. Scatter plots with fitted regression lines were used to explore bivariate relationships between lifestyle factors—such as *Age*, *Stress.Level*, and *Physical.Activity.Level*—and systolic blood pressure. A multiple linear regression model was developed to quantify how these predictors influenced systolic blood pressure. Additional analyses included Pearson correlation to examine associations among numerical variables, group comparisons of heart rate across gender and occupation, and conditional probability calculations to estimate the likelihood of having a Normal BMI for specific subgroups.

2.3 Analysis

The analysis began with an exploratory assessment of the numerical variables using visualizations to identify distribution patterns and potential outliers. As illustrated in **Figure 1(a)**, the boxplot of *Heart.Rate* reveals several unusually high values, indicating the presence of multiple outliers. This observation is supported by the histogram shown in **Figure 1(b)**, which displays a right-skewed distribution with a long tail extending beyond 78 bpm. These visuals confirm that the heart rate variable contains extreme values that influence its overall shape.

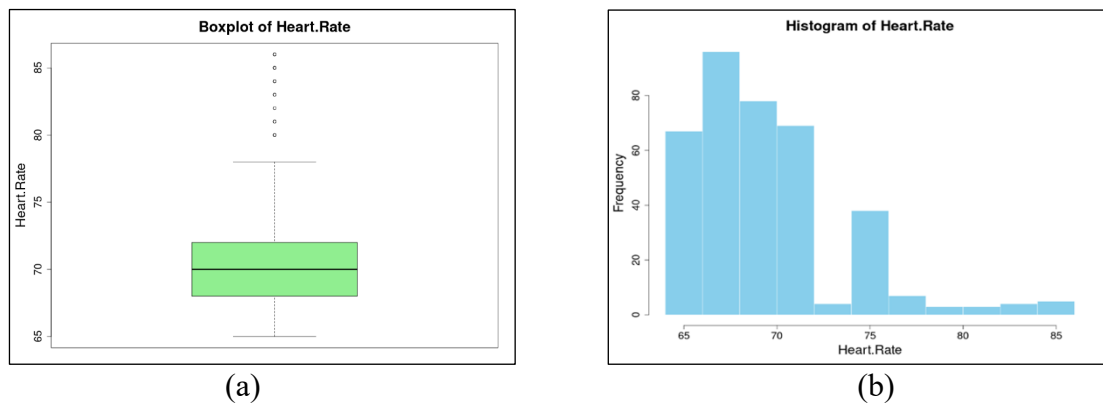


Figure 1. (a) Boxplot of Heart Rate, (b) Histogram of Heart Rate

Heart rate was further examined across demographic and occupational groups. **Figure 2(a)** presents a boxplot comparing *Heart.Rate* between males and females, showing substantial overlap in their distributions and suggesting minimal gender-based differences. In contrast, **Figure 2(b)** shows the mean heart rate across occupations, where sales-related roles exhibit noticeably higher average heart rates compared to healthcare, engineering, and teaching professions. This indicates that occupation appears to have a modest effect on heart rate variability within the dataset.

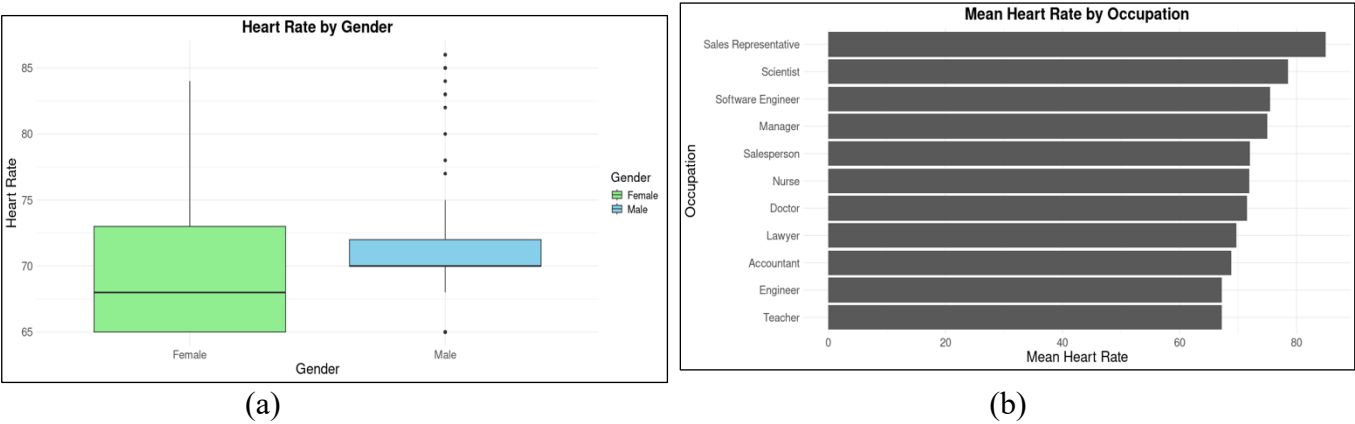


Figure 2. (a) Boxplot of Heart Rate by Gender, (b) Mean Heart Rate by Occupation

To assess relationships between heart rate and continuous lifestyle indicators, scatterplots with fitted regression lines were produced. **Figure 3(a)** shows a weak negative correlation between *Daily.Steps* and *Heart.Rate*, indicating that individuals who walk more steps tend to have slightly lower resting heart rates. Similarly, **Figure 3(b)** displays the relationship between *Age* and *Heart.Rate*, revealing another mild negative trend, suggesting that heart rate decreases slightly with increasing age. In both cases, the wide spread of data points around the regression line indicates that these associations are weak.

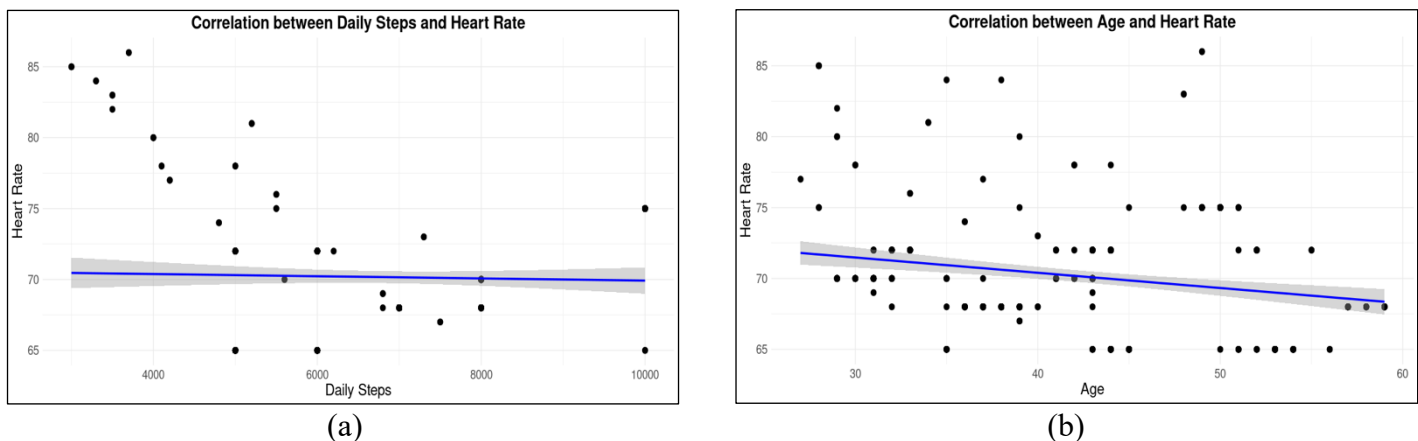


Figure 3. (a) Daily Steps vs. Heart Rate, (b) Age vs. Heart Rate.

To further examine key physiological relationships, a correlation heatmap was generated for *Sleep Disorder*, *Systolic Blood Pressure*, *Diastolic Blood Pressure*, and *Sleep Duration*, as shown in **Figure 4**. The heatmap highlights a very strong positive correlation between systolic and diastolic blood pressure ($r \approx 0.97$), consistent with expected physiological dependence. Sleep disorder, however, shows only weak associations with both blood pressure components and sleep duration, indicating limited linear relationships among these variables.

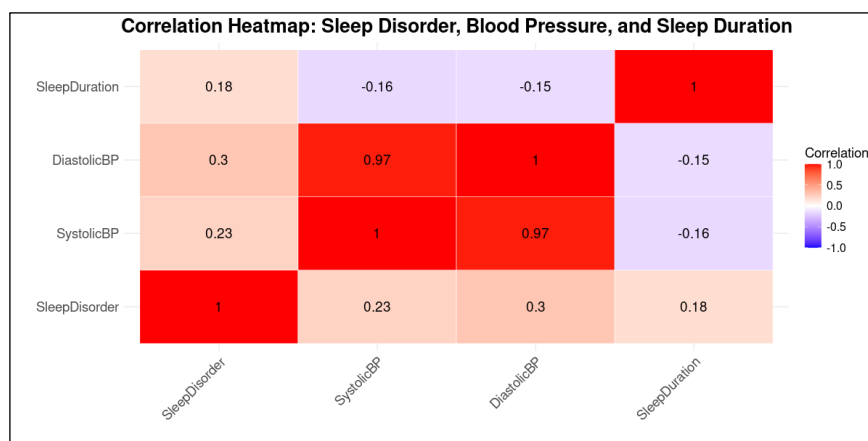


Figure 4. Correlation heatmap of key health variables

These visual analyses provided a foundational understanding of distribution patterns, inter-variable relationships, and subgroup differences. They also inform the interpretation of statistical findings presented in the Results section.

2.4 Results

The results of the statistical analysis revealed several notable patterns across lifestyle and physiological variables. As summarized in **Table 1**, both sleep duration and heart rate contained identifiable outliers, with three extreme sleep duration values (2, 12, and 15 hours) and fifteen unusually high heart rate readings contributing to right-skewed distributions. Age, stress level, and physical activity each demonstrated weak positive trends with systolic blood pressure, indicating that these factors had minimal predictive value. Gender differences in heart rate were negligible, while occupation showed moderate variation, with sales-related roles exhibiting the highest mean heart rate. Daily steps and age both showed weak negative correlations with heart rate, suggesting slightly lower heart rates among more active or older individuals. BMI results showed clear subgroup differences, with Doctors having a substantially higher probability of Normal BMI (88.7%) compared to males overall (67.7%). Correlation analysis revealed very strong dependence between systolic and diastolic blood pressure ($r = 0.97$), while sleep disorder showed only weak associations with both blood pressure and sleep duration. Overall, the findings in **Table 1** indicate that most relationships in the dataset are weak in magnitude, with only a few variables showing moderate or strong associations.

Table 1. Table, version 1

Analysis Component	Key Result	Interpretation
Sleep Duration Outliers	2, 12, 15 hours	Mild effect on mean; right-skewed distribution
Heart Rate Outliers	15 values	Contributed to distribution skewness
Age → Systolic BP	Weak positive trend	Older age slightly increases systolic BP
Stress Level → Systolic BP	Very weak positive trend	Minimal predictive value
Physical Activity → Systolic BP	Weak positive trend	Not clinically meaningful
Gender → Heart Rate	Weak difference	Overlapping male/female distributions
Occupation → Heart Rate	Moderate variation	Sales occupations showed highest HR
Daily Steps → Heart Rate	Weak negative ($r \approx -0.05$ to -0.1)	More steps slightly reduce HR
Age → Heart Rate	Weak negative	HR slightly declines with age
P(BMI = Normal Doctor)		88.7%
P(BMI = Normal Male)		67.7%
Sleep Disorder → BP	$r = 0.23-0.30$	Weak positive correlation
Sleep Disorder → Sleep Duration	$r = 0.18$	Very weak association
Systolic vs. Diastolic BP	$r = 0.97$	Extremely strong correlation

3. Conclusion

The findings of this study indicate that while several demographic and lifestyle factors—such as age, stress level, physical activity, daily steps, and gender—show observable trends with physiological indicators, their individual effects are generally weak. Outlier analysis identified a small number of extreme sleep duration and heart rate values, though their impact on overall distributions was limited. More notable patterns emerged in specific areas: occupation demonstrated moderate variation in heart rate, with sales-related roles showing the highest averages, and Doctors exhibited a substantially higher probability of having a Normal BMI compared to males overall. The strongest relationship in the dataset was the expected physiological correlation between systolic and diastolic blood pressure, which showed an exceptionally high correlation.

Overall, the results suggest that health outcomes in this dataset are influenced by multiple interacting factors rather than any single dominant predictor. At the same time, the study's conclusions are constrained by the small dataset size, weak correlations, and reliance on self-reported measures, which limit predictive strength and generalizability. These findings underscore both the complexity of lifestyle–health relationships and the need for broader, multi-dimensional analyses supported by larger and higher-quality datasets in future research.

References

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Appendix

Code

```
# 1. load dataset

df <- read.csv("healthy_lifestyle.csv", stringsAsFactors = FALSE)

#-----Handling missing values

# Show NA counts per column
na_counts <- colSums(is.na(df))
na_counts

# Show empty / whitespace-only string counts per column
empty_string_counts <- sapply(df, function(col) {
  if (is.factor(col)) col <- as.character(col)
  if (is.character(col)) {
    sum(trimws(col) == "" & !is.na(col))
  } else {
    0
  }
})
empty_string_counts

# Convert empty / whitespace-only strings to NA (for character columns)
char_cols <- sapply(df, is.character)

df[char_cols] <- lapply(df[char_cols], function(col) {
  col[trimws(col) == ""] <- NA
  col
})

# Check again after cleaning
colSums(is.na(df))

# 4. Remove any rows that contain NA in any column
df_clean <- na.omit(df)

#-----Descriptive statistics (age, occupation, sleep duration)

summary(df$Age) # shows Min, 1st Qu., Median, Mean, 3rd Qu., Max
sd(df$Age) # Standard deviation
getmode(df$Age) # Mode

# Frequency table
table(df$Occupation)

# Mode (most frequent category)
getmode(df$Occupation)

summary(df$Sleep.Duration) # Min, 1st Qu., Median, Mean, 3rd Qu., Max
sd(df$Sleep.Duration) # Standard deviation
getmode(df$Sleep.Duration) # Mode

#-----Visualizations for numeric values
```

```

# Select only numeric columns
numeric_cols <- sapply(df, is.numeric)
df_numeric <- df[, numeric_cols]

par(
  mar      = c(5, 5, 4, 2) + 0.1, # margins
  cex.main = 2.2, # title size
  cex.lab  = 2.0, # axis labels
  cex.axis = 1.6  # tick labels
)
# Histogram for each numeric column
for(col in names(df_numeric)) {
  hist(df_numeric[[col]],
       main = paste("Histogram of", col),
       xlab = col,
       col = "skyblue",
       border = "white")
}
par(
  mar      = c(5, 5, 4, 2) + 0.1,
  cex.main = 2.2,
  cex.lab  = 2.0,
  cex.axis = 1.6
)

colors()
# Boxplot for each numeric column to detect outliers
for(col in names(df_numeric)) {
  boxplot(df_numeric[[col]],
         main = paste("Boxplot of", col),
         ylab = col,
         col = "lightgreen")
}

library(e1071)
for(col in names(df_numeric)) {
  cat(col, "skewness:", skewness(df_numeric[[col]]), "\n")
}

#-----Effect of Age, Stress Level, and Physical Activity on Blood
Pressure

# Split BP into systolic and diastolic
bp_split <- strsplit(as.character(df$Blood.Pressure), "/")

# Extract systolic and diastolic as numeric variables
df$Systolic <- sapply(bp_split, function(x) as.numeric(x[1]))
df$Diastolic <- sapply(bp_split, function(x) as.numeric(x[2]))

# Multiple linear regression with Systolic BP as response
model <- lm(Systolic ~ Age + Stress.Level + Physical.Activity.Level, data
= df)

# Show regression output
summary(model)

```

```

#-----Correlation for heart rate with (gender, occupation, daily
steps, age) with graph

# Numeric correlations
cor(df$Daily.Steps, df$Heart.Rate, use = "complete.obs")
cor(df$Age, df$Heart.Rate, use = "complete.obs")

aggregate(Heart.Rate ~ Gender, data = df, mean)
aggregate(Heart.Rate ~ Occupation, data = df, mean)

#-Gender
library(ggplot2)

ggplot(df, aes(x = Gender, y = Heart.Rate, fill = Gender)) +
  geom_boxplot() +
  scale_fill_manual(values = c("Female" = "lightgreen",
                                "Male" = "skyblue")) +

  labs(
    title = "Heart Rate by Gender",
    x = "Gender",
    y = "Heart Rate"
  ) +
  theme_minimal() +
  theme(
    text = element_text(size = 16), #
overall bigger text
    plot.title = element_text(hjust = 0.5, size = 20, face = "bold"), #
centered big title
    axis.title = element_text(size = 18), # big
axis labels
    axis.text = element_text(size = 14) # big
tick labels
  )

#-Occupation
library(dplyr)

# Remove NA or empty occupations
df_occ <- df %>%
  mutate(Occupation = trimws(Occupation)) %>% # remove spaces
  filter(!is.na(Occupation), Occupation != "") # drop NA & empty

# Aggregate: mean Heart Rate by Occupation
occ_hr <- df_occ %>%
  group_by(Occupation) %>%
  summarise(mean_hr = mean(Heart.Rate, na.rm = TRUE)) %>%
  arrange(mean_hr) # sort by mean

ggplot(occ_hr, aes(x = reorder(Occupation, mean_hr), y = mean_hr)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Mean Heart Rate by Occupation",
    x = "Occupation",
    y = "Mean Heart Rate"
  ) +
  theme_minimal() +
  theme(
    text = element_text(size = 16), # overall text size

```

```

    plot.title = element_text(hjust = 0.5, size = 20, face = "bold"), #
center + bigger title
    axis.title = element_text(size = 18),
    axis.text = element_text(size = 14)
  )

#-Daily Steps
ggplot(df, aes(x = Daily.Steps, y = Heart.Rate)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", col = "blue", linewidth = 1.2) +
  labs(
    title = "Correlation between Daily Steps and Heart Rate",
    x = "Daily Steps",
    y = "Heart Rate"
  ) +
  theme_minimal() +
  theme(
    text = element_text(size = 16), #
bigger overall text
    plot.title = element_text(hjust = 0.5, size = 20, face = "bold"), #
centered big title
    axis.title = element_text(size = 18), #
bigger axis labels
    axis.text = element_text(size = 14) #
bigger tick mark labels
  )

#-Age
ggplot(df, aes(x = Age, y = Heart.Rate)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", col = "blue", linewidth = 1.2) +
  labs(
    title = "Correlation between Age and Heart Rate",
    x = "Age",
    y = "Heart Rate"
  ) +
  theme_minimal() +
  theme(
    text = element_text(size = 16), #
bigger overall text
    plot.title = element_text(hjust = 0.5, size = 20, face = "bold"), #
centered big title
    axis.title = element_text(size = 18), #
bigger axis labels
    axis.text = element_text(size = 14) #
bigger tick labels
  )

#-----Probability of normal BMI

# Probability that BMI = "Normal" given Occupation = "Doctor"
p_bmi_normal_given_doctor <- sum(df$BMI == "Normal" & df$Occupation ==
"Doctor", na.rm = TRUE) /
  sum(df$Occupation == "Doctor", na.rm = TRUE)

# Probability that BMI = "Normal" given Gender = "Male"
p_bmi_normal_given_male <- sum(df$BMI == "Normal" & df$Gender == "Male",
na.rm = TRUE) /
  sum(df$Gender == "Male", na.rm = TRUE)

```

```

# Display results
p_bmi_normal_given_doctor
p_bmi_normal_given_male

#-----Outlier detection for sleep duration

# Summary statistics
summary(df$Sleep.Duration)

# Identify outliers using the IQR method
Q1 <- quantile(df$Sleep.Duration, 0.25, na.rm = TRUE)
Q3 <- quantile(df$Sleep.Duration, 0.75, na.rm = TRUE)
IQR <- Q3 - Q1

lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

# Detect outlier values
outliers <- df$Sleep.Duration[df$Sleep.Duration < lower_bound |
df$Sleep.Duration > upper_bound]
outliers

# Effect of outliers: compare mean vs median
mean_sleep <- mean(df$Sleep.Duration, na.rm = TRUE)
median_sleep <- median(df$Sleep.Duration, na.rm = TRUE)

mean_sleep
median_sleep

#-----Correlation with sleep disorder

library(ggplot2)
library(reshape2)
library(dplyr)

# Convert Sleep Disorder to numeric (0/1/2 style based on factor levels)
df$SleepDisorder.Num <- as.numeric(as.factor(df$Sleep.Disorder))

# Extract systolic and diastolic from Blood Pressure (e.g., "120/80")
df$BloodPressure.Systolic <- as.numeric(sub("/.*", "",
df$Blood.Pressure))
df$BloodPressure.Diastolic <- as.numeric(sub(".*/", "",
df$Blood.Pressure))

# Sleep.Duration is already numeric

# Select variables for correlation
df.Corr <- df %>%
  select(SleepDisorder.Num,
         BloodPressure.Systolic,
         BloodPressure.Diastolic,
         Sleep.Duration)

# Optional: rename columns for nicer labels in the heatmap
colnames(df.Corr) <- c("SleepDisorder",
                      "SystolicBP",
                      "DiastolicBP",
                      "SleepDuration")

```

```

# Calculate correlation matrix
cor.Matrix <- cor(df.Corr, use = "complete.obs")
cor.Matrix

# Extract correlation of SleepDisorder with other variables (if needed)
cor.With.SleepDisorder <- cor.Matrix["SleepDisorder", ]
cor.With.SleepDisorder

# Melt correlation matrix for plotting
melted.Cor <- melt(cor.Matrix)

# Plot heatmap with bigger text and centered title
ggplot(data = melted.Cor, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(
    low = "blue", high = "red", mid = "white",
    midpoint = 0, limit = c(-1, 1), space = "Lab",
    name = "Correlation"
  ) +
  geom_text(aes(label = round(value, 2)), color = "black", size = 5) +
  labs(
    title = "Correlation Heatmap: Sleep Disorder, Blood Pressure, and
Sleep Duration",
    x = "",
    y = ""
  ) +
  theme_minimal() +
  theme(
    text = element_text(size = 16), # overall text size
    plot.title = element_text(hjust = 0.5, size = 20, face = "bold"), #
centered, bigger title
    axis.text.x = element_text(angle = 45, hjust = 1, size = 14),
    axis.text.y = element_text(size = 14),
    legend.title = element_text(size = 14),
    legend.text = element_text(size = 12)
  )

```